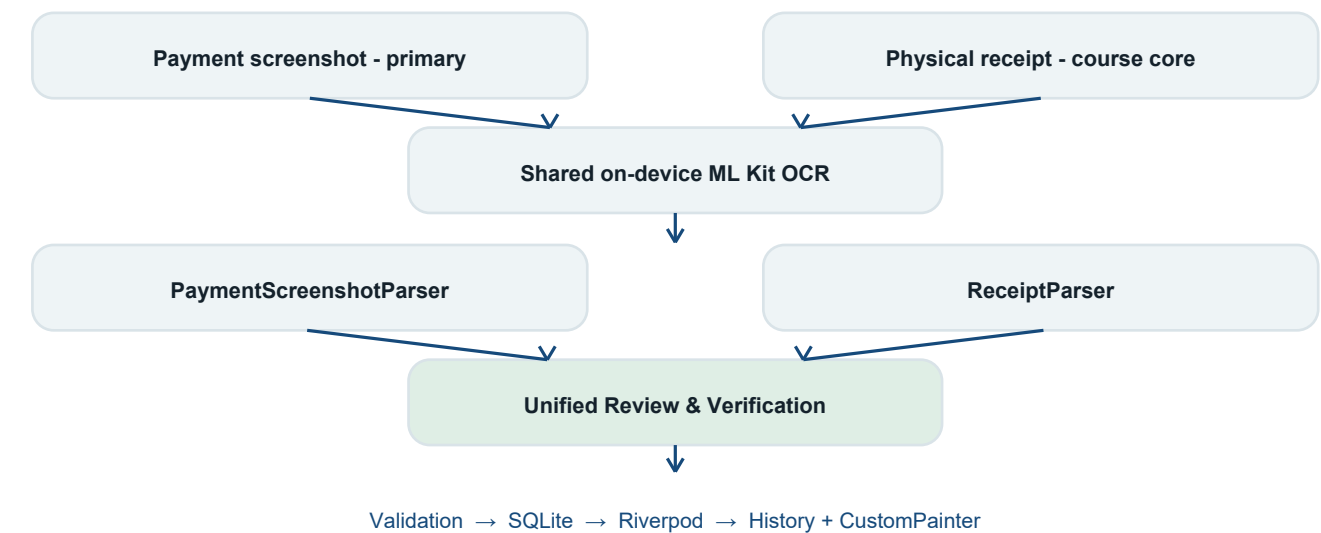


Smart Expense Capture & Tracker

Technical report · Mini-Project 3 · Flutter / Dart · October 2026

Vietnamese users often keep a bank or e-wallet payment screenshot rather than a paper receipt. The app therefore offers screenshot import as its primary entry, while retaining the course-required physical receipt camera and OCR workflow. Both inputs use on-device recognition and require human verification before local storage.

Architecture



Course core and product extension

Course required core	Product extension
Camera/gallery receipt, ML Kit OCR, receipt total/date/merchant heuristics, review, SQLite CRUD, Riverpod 2, custom charts.	Bank and e-wallet screenshot import, payment-specific parser, status, source label, optional provider/reference, duplicate warning.

Design: Material 3, a restrained blue/green palette, solid surfaces, 48 dp touch targets, light/dark themes, an original receipt/check brand mark, Vietnamese-first and English UI, and one review form adapted to the input source.

OCR heuristics and persistence

Recognition uses `google_mlkit_text_recognition` with the Latin model. The parsers apply Dart RegExp and ordered rules; uncertain fields stay blank. No OCR value is committed directly.

Field	Detection rule	Review fallback
Receipt total	Weighted total/tổng tiền/thanh toán/cộng tiền/amount due line; adjacent amount line; VND grouping.	Blank amount
Receipt merchant	First plausible header; reject invoice, address, phone and date labels.	Blank merchant
Date/time	Strict dd/MM/yyyy or yyyy-MM-dd, plus HH:mm for payments; reject impossible dates.	Editable date/time
Payment amount	Prefer số tiền/amount and currency-bearing lines; normalize 150000, 150.000, 150,000.	Blank amount
Recipient/status	Labelled recipient or cautious post-date name; detect failed/pending before success phrases.	Edit recipient; require success
Transfer note	Labelled content, transfer phrase, or text after a destination account number; never copy the number.	Optional editable note
Duplicate	Reference match, or same source + amount + date + recipient.	Warn; user may save anyway

Data and state

SQLite stores integer VND amounts, date/time, category, source, merchant/recipient, optional status/provider/reference/note, image path, and timestamps. Version 2 adds payment columns with an additive migration. ExpenseDatabase owns SQL; a Riverpod 2 AsyncNotifier refreshes history and analytics after each successful mutation.

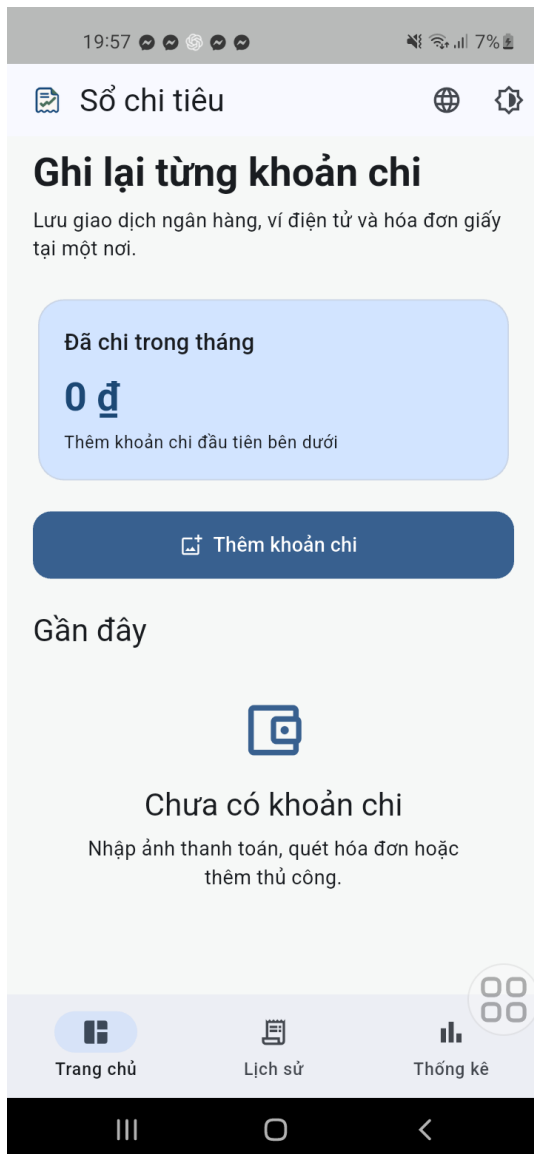
Privacy, testing, and limits

OCR and image processing stay on-device. Payment OCR text is visible during review but not persisted because it may contain account numbers; screenshots remain in app-owned storage. Unit/widget tests cover receipt and payment parsing, form correction/validation, SQLite CRUD, duplicate detection, and aggregation. The Android integration test mocks picker/ML Kit while exercising the real parsers and UI.

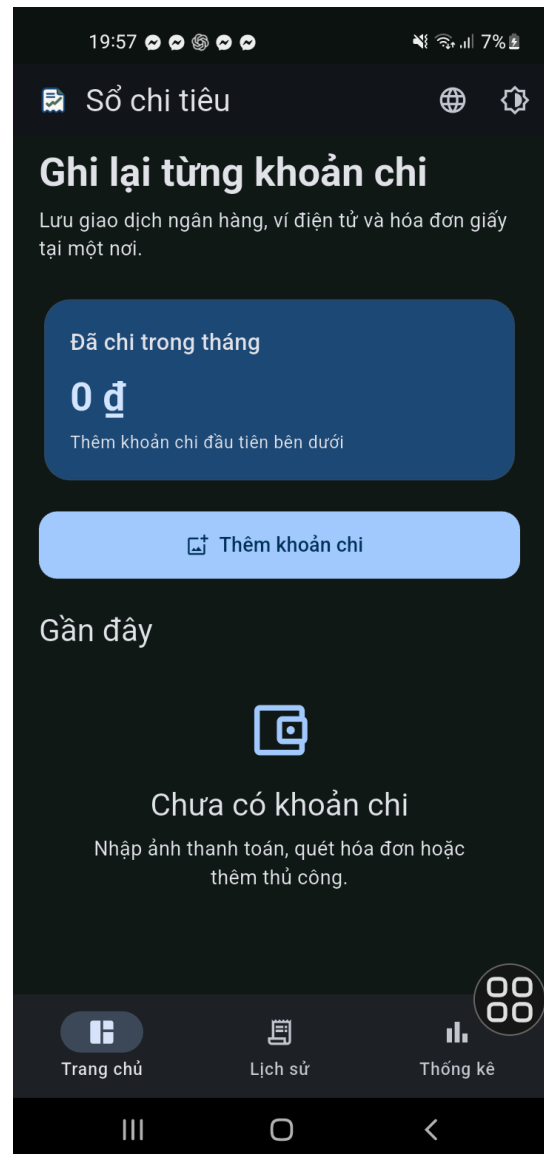
Limitations: noisy or handwritten inputs need correction; ambiguous VND decimals are not guessed; bank layouts vary; duplicate matching is advisory. Release OCR was exercised with three payment screenshots on a Samsung Android phone; camera receipt quality still needs a real paper receipt. iOS build validation requires macOS and Xcode.

Material 3 interface

These screenshots show the signed Android release on a physical Samsung phone. The Vietnamese interface, original receipt mark, light theme, and dark theme are visible. The empty state reflects a fresh local database; private payment images are omitted.



Light mode



Dark mode

Demo sequence

Open app → scan a real receipt → show OCR text and Regex-derived fields → correct in Review & Verification → save → history → animated category donut and weekly bars. Then import a bank or wallet screenshot, verify status/recipient/amount, and show the combined analytics update.

Implementation references: repository PRD and README; Flutter documentation; google_mlkit_text_recognition and image_picker package documentation on pub.dev.